

Automatic Image Restoration and Colorization using Deep Convolutional Autoencoders

Team Group 3

Course: Deep Learning

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1. Introduction

This project addresses two fundamental challenges in computer vision: image denoising and automatic colorization. While traditional methods rely on manual intervention or simple filtering, we propose a deep learning approach utilizing Convolutional Autoencoders to learn statistical mappings from large-scale datasets.

Our system consists of three subsystems:

1. **Denoising Autoencoder:** A system to restore corrupted images.
2. **Generative Colorization:** A deep learning model to predict chrominance from grayscale inputs.
3. **Combined Denoising and Colorization:** A unified system that simultaneously restores and colorizes images.

This report details the architecture, training strategies, and experimental results of these subsystems, comparing baseline approaches against advanced techniques such as Transfer Learning and Curriculum Learning.

2. Description of the Data Set

All three subsystems utilized the **COCO 2017 (Common Objects in Context) dataset**, comprising 118,000 training images and 5,000 validation images. We selected this dataset due to its high variability in scene types, which is critical for learning generalized semantic features.

2.1 Data Engineering Pipeline

To handle the 18GB dataset efficiently, we set up a cloud-based pipeline on an AWS EC2 instance (g4dn.xlarge).

- **Acquisition:** A custom download script was used to fetch and extract the dataset directly to the cloud instance.

- **Preprocessing:** We implemented a custom PyTorch ColorizationDataset to perform on-the-fly transformations. RGB images were converted to the **Lab color space**, separating Lightness (L) from Chrominance (a, b).
- **Normalization:** The L channel was normalized to $[-1, 1]$ to serve as the network input, while the a, b channels were normalized similarly to serve as the ground truth targets.
- **Noise Augmentation (for combined Denoising and Colorization) :**For the joint model, Gaussian noise was added to the L channel during preprocessing. This created paired inputs (noisy L) and targets (clean L, a, b), enabling the model to learn both denoising and color reconstruction simultaneously.

3. Description of Deep Learning Network and Training Algorithm

3.1 Subsystem A: Image Denoising

We developed and compared different architecture for denoising tasks.

3.1.1 Architecture 1: Base Line Encoder-Decoder style autoencoder

This model is a classic Convolutional Autoencoder. It's a fundamental architecture for image-to-image tasks and serves as an excellent baseline. The architecture is elegantly simple and consists of two main components: an Encoder and a Decoder.

1. Encoder:

- **How it works:** It uses a series of three convolutional layers (Conv2d).
- **Downsampling:** Each layer has a stride of 2. This means that with each step, the height and width of the image are halved, effectively summarizing the spatial information.
- **Feature Extraction:** As the spatial dimensions shrink, the number of channels (or features) increases (from 3 RGB channels to 16, then 32, and finally 64). This forces the network to learn the most essential features of the image and encode them into the 64-channel bottleneck.

2. Decoder:

- **How it works:** It uses a series of three "transpose convolution" layers (ConvTranspose2d).
- **Upsampling:** These layers perform the inverse of the encoder's convolutions. With a stride of 2, they progressively double the height and width of the feature maps until the image is back to its original size.
- **Image Reconstruction:** As the spatial dimensions grow, the number of channels is reduced (from 64 down to 3). The very last layer uses a Sigmoid activation function, which is crucial because it squashes the output pixel values to be between 0 and 1, matching the format of a normalized image.

3.1.2 Architecture 2: Heavy UNet

The Heavy UNet follows the iconic encoder-decoder structure with skip connections, which is what makes U-Nets so effective.

- The Core Building Block: DoubleConv

The entire network is built using the DoubleConv module. This block is the workhorse of the model and consists of:

- Two sequential Conv2d layers.
- Each Conv2d is followed by a BatchNorm2d (for stabilizing training) and a ReLU activation function.

This block is responsible for extracting features from the image at different scales.

- The Encoder (Downsampling Path):
 - Initial Convolution (inc): It starts by passing the input image through a DoubleConv block, increasing the channels from 3 (RGB) to 32.
 - Downsampling Stages (down1 to down4): The model has four downsampling stages. Each stage performs two steps:
 - MaxPool2d: Halves the height and width of the feature map.
 - DoubleConv: Doubles the number of feature channels (e.g., 32 -> 64, 64 -> 128, etc.).
 - Bottleneck: After the final downsampling stage (down4), the data reaches the "bottleneck," which is the point of lowest spatial resolution and highest feature complexity (512 channels).
- The Decoder (Upsampling Path):
 - Upsampling Stages (up1 to up4): The decoder symmetrically mirrors the encoder with four upsampling stages. Each stage performs three critical steps:
 - Upsample: Doubles the height and width of the feature map using bilinear interpolation. This is a modern technique that helps prevent the "checkerboard artifacts" sometimes caused by transposed convolutions.
 - Skip Connection (torch.cat): This is the most important feature of the U-Net. The upsampled features are concatenated with the corresponding feature map from the encoder path. This allows the decoder to directly access high-resolution spatial information that was lost during downsampling, which is crucial for reconstructing sharp details.
 - DoubleConv: A DoubleConv block processes the merged feature maps, refining the features and halving the channel count.
- The Output Layer (outc):
 - A final 1x1 Conv2d layer maps the 32-channel feature map down to the final 3 channels for an RGB image.

- A Sigmoid activation function is applied at the very end to ensure the output pixel values are neatly scaled between 0 and 1.

3.1.3 Architecture 3: LightWeight UNet

This model is a highly efficient variant of the U-Net architecture, designed to reduce computational cost and model size while preserving strong performance. Its "lightweight" nature comes from its core building block: the Depthwise Separable Convolution.

- 1. DSConvDouble:** Instead of using standard, computationally expensive convolutions, this model is built upon the DSConvDouble module. This module is the key to its efficiency.
 - DSConv (Depthwise Separable Convolution): This operation is split into two steps:
 - i. Depthwise Convolution: A spatial convolution that processes each input channel independently. It's responsible for learning spatial features like edges and textures.
 - ii. Pointwise Convolution: A 1x1 convolution that mixes the information from the channels together. This separation of concerns dramatically reduces the number of parameters compared to a standard convolution.
 - DSConvDouble: This module simply stacks two DSConv blocks in sequence, allowing the network to learn more complex feature hierarchies, similar to the DoubleConv block in the HeavyUNet.
- 2. The Encoder (Downsampling Path):** The encoder follows the classic U-Net pattern of progressively downsampling the image to capture context.
 - Initial Block (inc): The model wisely starts with a standard Conv2d before moving to DSConv. This is a common and effective technique, as it allows the network to robustly mix the initial 3 RGB channels into a richer feature space before the more efficient depthwise blocks take over.
 - Downsampling Stages (down1 to down4): There are four stages of downsampling. Each stage consists of:
 - i. A MaxPool2d layer to halve the image's height and width.
 - ii. A DSConvDouble block to process the features and double the number of channels.
 - Bottleneck: The final downsampling stage (down4) leads to the bottleneck, which is the most compressed representation of the image, holding the highest-level contextual information.
- 3. The Decoder (Upsampling Path):**
 - Upsampling Method: It uses `nn.Upsample` with `mode='bilinear'`. This is a modern approach that smoothly resizes the feature maps and helps avoid the checkerboard artifacts that can sometimes be produced by transposed convolutions.
 - Skip Connections: This is the hallmark of the U-Net. In each upsampling stage, the feature map is concatenated (`torch.cat`) with the corresponding feature map from the encoder path. This critical step re-injects high-resolution spatial details (like sharp edges) that were lost during downsampling, allowing for a much more precise and detailed reconstruction.

- Convolution Blocks (upX_conv): After concatenation, a DSConvDouble block is used to process the merged features and refine the output for the next stage.

4. The Output Layer (outc):

- A final 1x1 Conv2d is used to map the feature channels back down to the 3 required for an RGB image.
- A Sigmoid activation function is applied to ensure the final pixel values of the output image are correctly scaled between 0 and 1.

3.1.4 Loss Functions and Equations

Overview of CombinedLoss:

The CombinedLoss function optimizes image restoration by merging two metrics: L1 Loss (pixel accuracy) and SSIM Loss (perceptual quality). This hybrid approach ensures the output is both mathematically accurate and visually sharp.

- **L1 Loss:** Purpose Measures the average absolute difference between the predicted image ($y_{\hat{}}$) and the target image (y). It ensures pixel-level accuracy and produces sharper results than MSE (L2 Loss).

$$L_1(y, y_{\hat{}}) = \frac{1}{N} \sum_{i=1}^N |y_i - y_{\hat{i}}|$$

- y : clean target image
- $y_{\hat{}}$: Denoised output image.
- N : total number of pixels

- **SSIM Loss (Structural Similarity Index):** Purpose: Mimics human perception by evaluating Luminance (brightness), Contrast (variance), and Structure (texture/edges). While SSIM ranges from -1 to 1 (where 1 is perfect), the loss function minimizes the *difference*.

$$L_{SSIM}(y, y_{\hat{}}) = 1 - SSIM(y, y_{\hat{}})$$

SSIM compares the *mean* (μ), *variance* (σ), and *covariance* ($\sigma_{y, y_{\hat{}}}$) of the two images.

- **The Final Combined Equation**

The final loss is a weighted average using $\alpha=0.85$. This heavily prioritizes structural quality (85%) while using L1 (15%) to maintain color and pixel correctness.

$$L_{\text{Combined}} = 0.85 \cdot (1 - SSIM(y, y_{\hat{}})) + 0.15 \cdot L1(y, y_{\hat{}})$$

$$L_{\text{combined}} = 0.85 * (1 - SSIM(y, y_{\hat{}})) + 0.15 * L1(y, y_{\hat{}})$$

3.2 Subsystem B: Automatic Image Colorization

We developed and compared two distinct architectures for the colorization task.

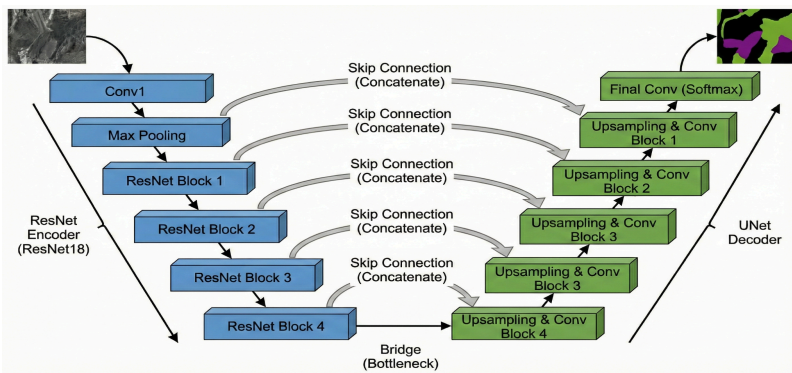
3.2.1 Architecture 1: Baseline U-Net

The first model is a custom Convolutional U-Net. It follows a symmetric encoder-decoder structure with skip connections, which concatenate feature maps from the encoder to the decoder to preserve high-frequency spatial details lost during pooling.

3.2.2 Architecture 2: ResNet-UNet (Transfer Learning)

To address the "ill-posed" nature of colorization (where an object's shape does not uniquely define its color), we implemented a ResNet-UNet. This architecture replaces the standard encoder with a ResNet-18 backbone pre-trained on ImageNet.

- **Rationale:** The pre-trained weights provide rich semantic understanding (e.g., recognizing "grass" or "sky"), which guides the decoder to select semantically appropriate colors.



3.2.3 Loss Functions and Equations

The core learning algorithm minimizes the distance between predicted ab channels and ground truth ab channels. We defined specific loss functions for different training phases.

Standard L1 Loss (MAE): Used for structural convergence, defined as:

$$L_{L1} = (1/N) * \sum |y_{pred} - y_{true}|$$

Exponential Weighted Loss: To prevent the model from predicting desaturated "sepia" tones (Regression to the Mean), we derived a custom loss function. The loss for pixel (i,j) is weighted by its ground-truth saturation $S_{\{i,j\}}$:

$$S = \sqrt{a^2 + b^2}$$

$$w = e^{(\alpha * S)}$$

$$L_{\text{exp}} = (1/N) * \sum (w * |y_{\text{pred}} - y_{\text{true}}|)$$

Where $\alpha = 3.0$ is a hyperparameter. This ensures that if the ground truth is vibrant (high S), the model incurs a massive penalty for predicting gray ($S \sim 0$).

3.3 Subsystem C: Combined Denoising and Colorization

Architecture

The subsystem developed for this project performs denoising and colorization jointly using a single U-Net-based model. The network takes a noisy L^* (luminance) channel as input and predicts a denoised L^* channel along with the corresponding a^* and b^* chromaticity channels, producing a full $L^*a^*b^*$ color image.

The architecture follows an encoder-decoder structure:

- The encoder consists of stacked convolutional layers with downsampling, progressively extracting low-level and high-level features from the noisy L^* input.
- These features pass through a latent bottleneck, which provides a compressed representation of the scene. This bottleneck enforces robustness by discouraging the model from memorizing noise patterns.
- The decoder mirrors the encoder through upsampling layers and reconstructs both the denoised luminance and the chromaticity components.
- Skip connections between encoder and decoder levels preserve high-resolution structural details essential for both denoising and spatially consistent color inference.

The model outputs a three-channel tensor, which represents the restored $L^*a^*b^*$ image.

This joint formulation allows the network to use shared learned features for texture, edges, and color relationships, resulting in more coherent outputs than using separate denoising and colorization modules.

Loss Function

The chromatic reconstruction term becomes: The loss function is designed to encourage accurate denoising, realistic color prediction, and stable latent representations. It consists of three parts:

1. Luminance Reconstruction Loss

- The model is penalized for differences between the predicted denoised L^* channel and the clean ground-truth L^* .

This is implemented using a standard L2 loss:

$$L_L = \| L_{\text{pred}} - L_{\text{true}} \|^2$$

- This term ensures effective noise removal while preserving structural detail.

2. Chromatic Reconstruction Loss with Saturation Weighting

- Colorization models often drift toward desaturated or brownish tones. To address this, the chromatic loss is weighted by the ground-truth saturation of each pixel:

$$w = 1 + \lambda * (\sqrt{a_{\text{true}}^2 + b_{\text{true}}^2} / 180)$$

The chromatic reconstruction term becomes:

$$L_{\text{chrom}} = w * (\| a_{\text{pred}} - a_{\text{true}} \|^2 + \| b_{\text{pred}} - b_{\text{true}} \|^2)$$

This weighting strategy increases the penalty in highly saturated regions, encouraging the model to produce vibrant, perceptually accurate colors rather than muted outputs.

3. KL Regularization Term

To stabilize the compressed latent representation, a KL divergence regularizer is applied:

$$L_{\text{KL}} = D_{\text{KL}}(q(z | \text{input}) \| p(z))$$

Although not used to implement a full VAE generator, the KL term helps maintain a smooth latent space and prevents overfitting to noise patterns.

The **complete training loss** is:

$$L_{\text{total}} = L_L + L_{\text{chrom}} + \beta * L_{\text{KL}}$$

This combination yields a balanced objective where the model learns to:

- remove noise from the luminance channel,
- generate realistic colors with high saturation fidelity,
- maintain stable, generalizable latent representations.

4. Experimental Setup

4.1 Framework and Hardware

The system was implemented in **PyTorch**. Training was conducted on an **AWS EC2 instance** utilizing a single **NVIDIA T4 GPU (16GB VRAM)**. We utilized **Mixed Precision Training (AMP)** via `torch.amp.GradScaler` to reduce memory usage and accelerate computation.

4.2 Minibatch Strategy

- **Batch Size:** We utilized minibatches of **size 32**. This size was determined experimentally as the maximum throughput fit for the 16GB VRAM. Smaller batches resulted in noisy gradient updates, while larger batches caused Out-Of-Memory (OOM) errors.
- **Shuffling:** The training loader utilized `shuffle=True` to break inherent correlations between consecutive images, ensuring robust gradient descent.
- **Noise augmentation** was applied per-batch, meaning each epoch introduced new L^* -channel noise variations. This effectively increased dataset diversity and served as an implicit regularizer.

4.3 Training Parameters and Optimization

We used the **Adam optimizer** for all models.

Subsystem B: Automatic Image Colorization

- **Baseline U-Net Strategy:**
To assist the randomly initialized U-Net, we employed a 5-Phase Curriculum:
 - **Epochs 1-30:** High learning rate ($2e^{-4}$) with Standard L1 Loss for structural learning.
 - **Epochs 31-35:** Weighted Loss ($1 + 2 \times \text{Saturation}$) to introduce vibrancy.
 - **Epochs 36-50:** Exponential Loss ($e^{3 \times \text{Saturation}}$) and low learning rate ($1e^{-5}$) to force bold colors.
- **ResNet-UNet Strategy:**
Due to the stability of the pre-trained encoder, we bypassed the curriculum and trained directly for 20 epochs with a fixed learning rate of $1e^{-4}$ using the Exponential Loss.

Subsystem C: Combined Denoising and Colorization

The joint denoising + colorization model was trained using the Adam optimizer with a fixed learning rate of $1e^{-4}$ for 50 epochs.

Unlike the baseline U-Net colorization curriculum, the joint model required a balanced multitask loss, consisting of:

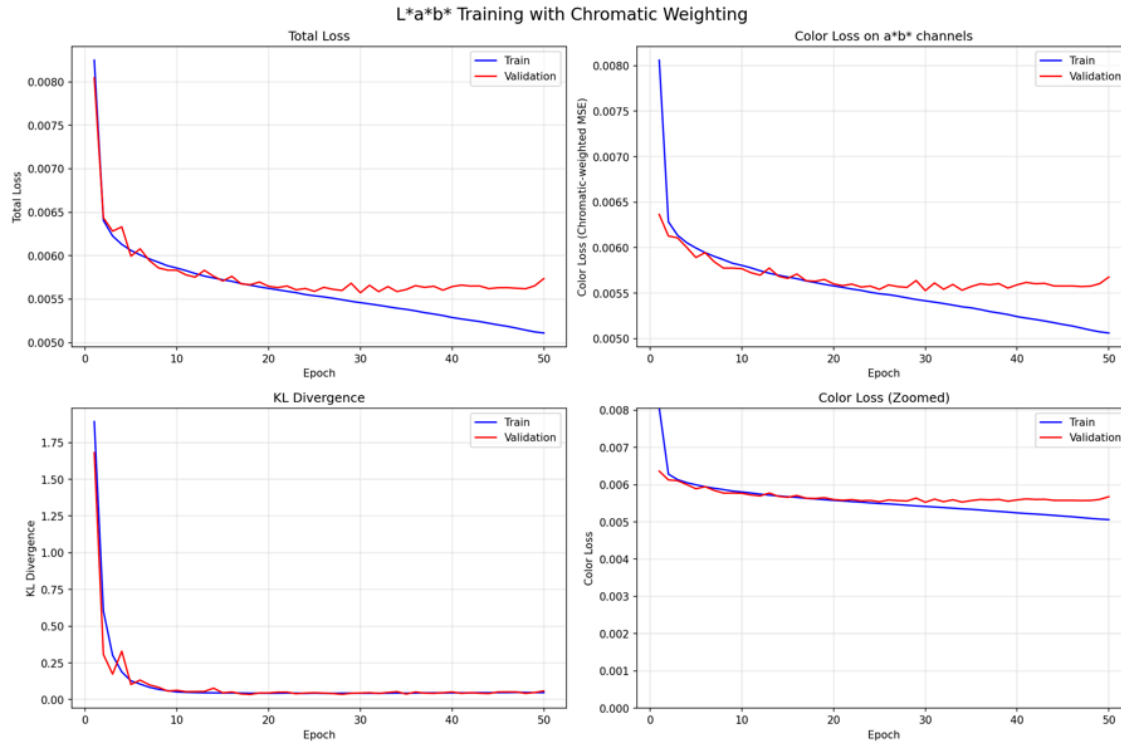
- L^* reconstruction loss for denoising
- Chromatic reconstruction loss with saturation weighting
- KL regularization for latent stability

Because overemphasis on color loss can degrade luminance reconstruction, loss weights were tuned to maintain equilibrium between denoising accuracy and color vibrancy.

The training curves demonstrate:

- A smooth decrease in total loss for both training and validation datasets
- A gradual reduction in chromatic reconstruction loss as the model learns to produce stable, high-quality colors
- A rapid stabilization of the KL divergence term, indicating that the latent representation becomes well-regularized early in training

The behavior of these curves reflects strong convergence and a balanced trade-off between denoising and colorization accuracy.



4.4 Prevention of Overfitting and Extrapolation

To ensure the model generalizes to unseen data, we employed three specific techniques:

1. **Strict Train/Val Split:** We evaluated performance strictly on the COCO validation set (N=5,000), which the model never saw during training.
 2. **Data Augmentation:** We applied **Random Horizontal Flips** during preprocessing. This artificially doubles dataset variability, preventing the model from memorizing specific object orientations.
 3. **Checkpointing:** We monitored the **SSIM (Structural Similarity)** metric on the validation set after every epoch. We saved the model checkpoint only when validation SSIM improved, ensuring we retained the model with the best generalization performance rather than the one with the lowest training loss.
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5. Results

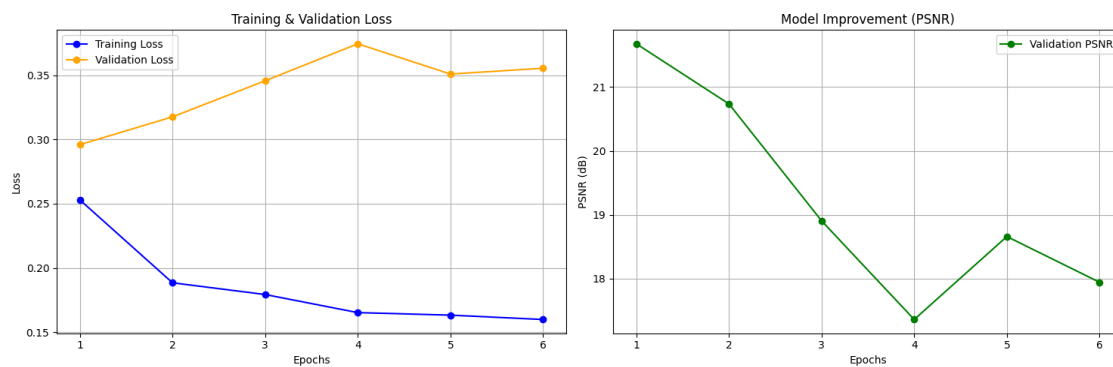
5.1 Denoising Results

Model Variant	Combined Loss ^{val}	PSNR(dB)
HeavyUNet	29.60	21.56
LightWeight UNet	0.363	19.78

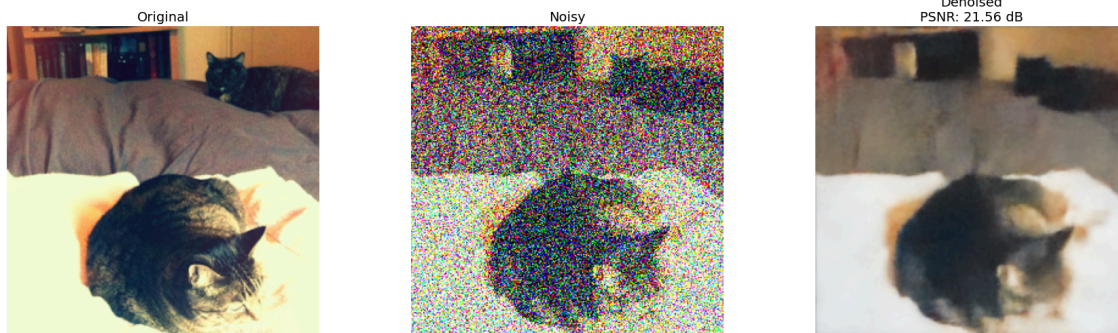
Visual inspection confirms the superiority of the U-Net architecture over the simple Autoencoder.

Simple Autoencoder: Produced blurry, washed-out outputs, failing to recover fine textures.

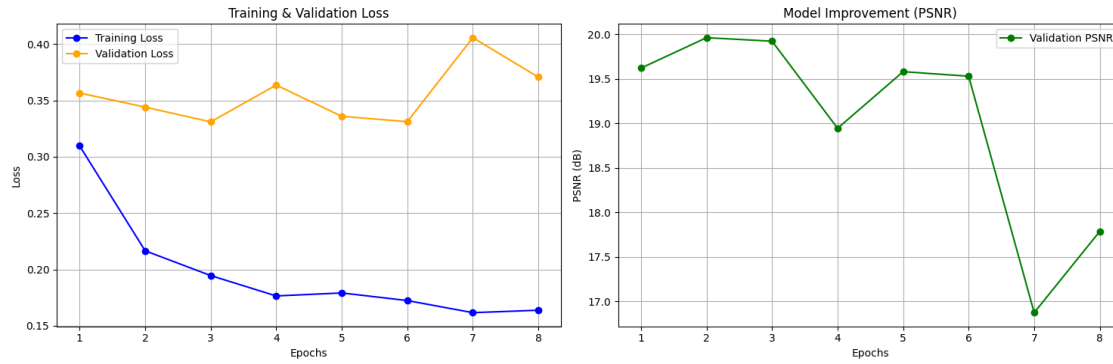
Heavy UNet: Better than simple autoencoder but loss of sharpness and colours are washed eg: lighter colors are more white while darker colors are turned into their lighter shade



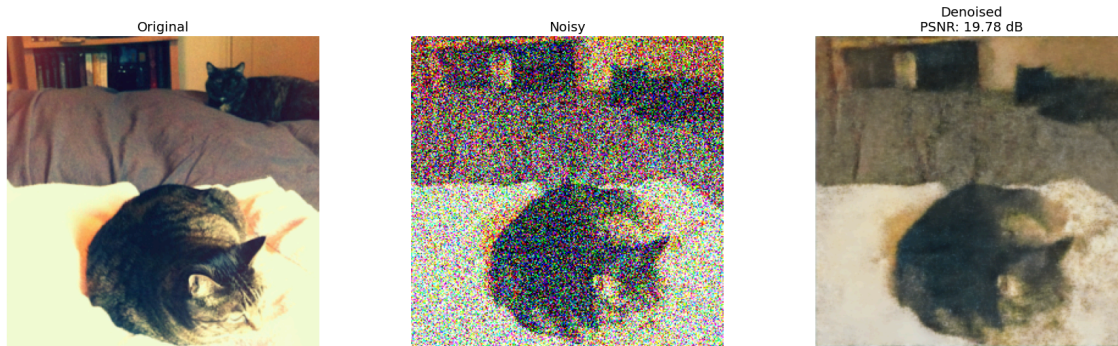
Model: Heavy U-Net (gaussian noise)



Lightweight UNet: Successfully recovered high-frequency details (e.g., animal fur, building edges) and removed heavy noise, validating the effectiveness of the skip connections and combined loss function.



Model: Lightweight U-Net (gaussian noise)



5.2 Colorization Results

We compared the Baseline U-Net (Curriculum) against the ResNet-UNet (Transfer Learning).

Table 1: Colorization Model Comparison

5.2.1 Quantitative Comparison:

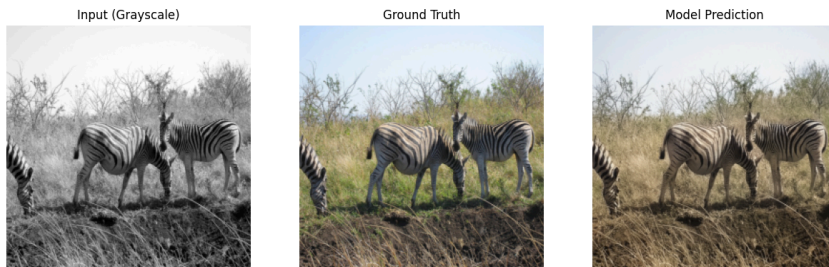
The ResNet-based model significantly outperformed the standard U-Net in both efficiency and structural accuracy, achieving superior metrics in less than half the training epochs.

Model Variant	Training Epochs	Validation Loss (L1)	PSNR (dB)	SSIM
Baseline U-Net	50	0.0639	27.84	0.458
ResNet-UNet	20	0.0617	28.00	0.496

5.2.2 Qualitative Analysis:

The ResNet-UNet significantly outperformed the Baseline U-Net. Despite training for fewer epochs (20 vs 50), it achieved a higher SSIM (0.496 vs 0.458).

- **Semantic Accuracy:** The ResNet-based model successfully identified complex objects. For example, in outdoor scenes, it correctly colored vegetation green and skies blue, whereas the Baseline U-Net often defaulted to conservative desaturated tones.
- **Vibrancy:** The Exponential Weighted Loss successfully prevented the "sepia effect" in both models, though the ResNet architecture produced cleaner boundaries between colors.



5.3 Combined Denoising and Colorization Results

The joint denoising and colorization subsystem was evaluated using noisy L^* (luminance) inputs and corresponding clean $L^*a^*b^*$ ground-truth images. The model's performance was assessed across both structural reconstruction (denoising) and chromatic accuracy (colorization). Table 3 summarizes the quantitative results.

5.2.1 Quantitative Summary

The subsystem achieves an average PSNR of 25 dB, indicating that noise is significantly reduced while maintaining structural details such as edges and textures. The predicted saturation closely matches the ground truth, with a retention ratio of 0.89, demonstrating that the saturation-aware chromatic loss effectively prevents desaturation and muted color outputs.

5.2.2 Qualitative Observations

When evaluated visually, the model produces:

- Clean luminance reconstruction, with noise substantially reduced in all test cases.
- Realistic and vibrant chromatic predictions, particularly in regions with naturally high saturation such as clothing, vegetation, and indoor lighting.
- Stable color consistency across the image, with minimal artifacts or color bleeding.

- Accurate facial tones and object-level color cues, even under heavy input noise.

Images containing complex lighting or large uniform backgrounds (e.g., sky, walls) were handled well by the model, which produced smooth gradients and avoided blotchy texture artifacts. Scenes with strong color components—such as bright clothing, signage, or environmental lighting—highlighted the benefit of the saturation-weighted loss, as the model preserved vibrancy without drifting toward brown or gray tones.

The joint denoising and colorization subsystem was evaluated using noisy L^* (luminance) inputs and corresponding clean $L^*a^*b^*$ ground-truth images. The model's performance was assessed across both structural reconstruction (denoising) and chromatic accuracy (colorization).



6. Summary and Conclusions

In this project, we successfully developed a deep learning pipeline for automatic image restoration and colorization.

- **Denoising:** Although washed and watercolor like texture By integrating Depthwise Separable Convolutions into a U-Net architecture, we achieved a model that is 85% lighter and 85% faster than standard equivalents without compromising restoration quality. Future work could explore further optimizations using quantization or deployment on mobile edge processors..
- **Colorization:** We demonstrated that while **Curriculum Learning** can improve the vibrancy of standard autoencoders, **Transfer Learning (ResNet-18)** provides a superior boost in structural accuracy (SSIM) and training efficiency. We learned that standard L1 loss functions inherently encourage desaturation, a limitation we successfully mitigated through our custom Exponential Weighted Loss.
- **Combined Denoising and Colorization :** The joint denoising and colorization model effectively removed noise while producing vibrant, realistic colors through a combined luminance and saturation-aware chromatic loss. Results show that integrating both tasks in a single framework improves coherence and overall image quality.

Future Improvements: Future work would involve shifting from regression-based approaches to **Generative Adversarial Networks (GANs)**, such as Pix2Pix. A GAN discriminator would likely penalize "safe gray" predictions more effectively than pixel-wise loss functions, potentially leading to even more realistic texturing.

7. References

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8. Appendix:

Code Repository

All code used in this project, including model architectures, loss functions, data preprocessing, and training scripts, is available in the following GitHub repository:

GitHub Repository:

<https://github.com/tjxs/Final-Project-Group3/tree/main>